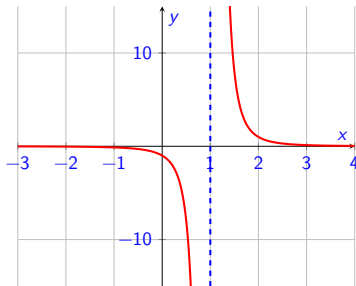


Week 03, Lecture 2 Vertical Asymptotes and Continuity

Dr Niall Madden

University of Galway

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This slides are by Niall Madden, with some content based on notes by Dr Kirsten Pfeiffer, and from Strang & Herman's "Calculus".

Reminders

- ▶ **Problem Set 1** due 5pm, Monday 28 September. You may access it multiple times, by clicking on [Assignments ... Problem Set 1 ...](#) and then, at the bottom of the page:
[Load Problem Set 1 in a new window](#)
- ▶ The **Tutorial Sheet** is available from Canvas.
- ▶ Problem Set 2 will open by Friday.
- ▶ The first (of two) class tests will take place Tuesday, 6th October.
If you wish to avail of Reasonable Accommodations for it, please complete this form:
<https://forms.cloud.microsoft/e/rBiq8RHYY2> by **10am, Tuesday 29 September**.

Today - a one-sided view of the following topics:

- 1 Infinite Limits
- 2 Digression: How fast can an object travel?
- 3 One-sided Limits
 - Notation
 - Piecewise functions
- Empty and full circle notation
- Existence of a limit
- 4 Vertical Asymptotes
 - Horizontal Asymptotes
- 5 Continuity
- 6 Exercises

For more, see Chapter 2 (Limits) in **Calculus** by Strang & Herman. See openstax.org/books/calculus-volume-1/pages/2-introduction

In particular §2.2: One-sided limits (§2.2.4+§2.2.5) and vertical asymptotes (§2.2.7).

Infinite Limits

So far, we've had lots of examples that are a little like:

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2 - x + 1}{(x - 1)^2} = 2.$$

(Check that this is correct).

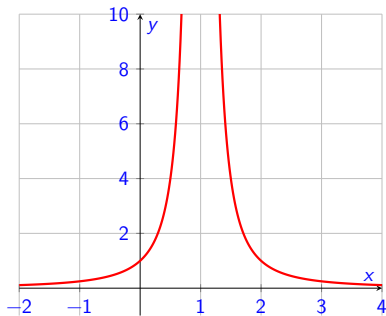
But what about

$$\lim_{x \rightarrow 1} \frac{1}{(x - 1)^2} = ???$$

Let's plot it and see:

Infinite Limits

$$f(x) = \frac{1}{(x-1)^2}$$



As x get closer and closer to 1, the value of $f(x)$ gets larger and larger. In fact, it becomes infinite.

For this we write

$$\lim_{x \rightarrow 1} f(x) = \infty.$$

Digression: How fast can an object travel?

- ▶ Q: Is there any limit to the speed at which an object can travel?
- ▶ A: Yes! (Assuming you believe Einstein)

Thanks to Einstein ($E = mc^2$), Lorenz and others, it is known that the mass of a moving charged particle behaves like

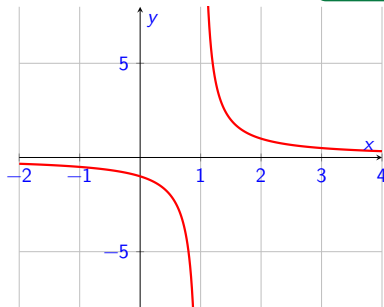
$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where m_0 is its mass at rest, c is the speed of light, and v is the particles current speed. What happens as $v \rightarrow c$?

One-sided Limits

Let's consider a motivating example, very similar to the one where we introduced ∞ .

$$f(x) = \frac{1}{x-1}$$



As x get closer and closer to 1, then $f(x) \rightarrow -\infty$ or $f(x) \rightarrow \infty$, depending on whether x approaches 1 from the left or right.

To express this, we need the concept of a **one-sided limit**

$\lim_{x \rightarrow a^-} f(x)$ is: **limit of f as x approaches a from the left**

$\lim_{x \rightarrow a^+} f(x)$ is: **limit of f as x approaches a from the right**

These mean that

- ▶ if $\lim_{x \rightarrow a^-} f(x) = L$, then we can make $f(x)$ as close to L as we would like by taking $x < a$ as close to a as needed.
- ▶ If $\lim_{x \rightarrow a^+} f(x) = L$, then we can make $f(x)$ as close to L as we would like by taking $x > a$ as close to a as needed.

$\lim_{x \rightarrow a^-} f(x)$ is: **limit of f as x approaches a from the left**

$\lim_{x \rightarrow a^+} f(x)$ is: **limit of f as x approaches a from the right**

In the previous example, with $f(x) = \frac{1}{x-1}$, we have

► $\lim_{x \rightarrow 1^-} f(x) = -\infty$

► $\lim_{x \rightarrow 1^+} f(x) = \infty$

In many important examples, we encounter functions that have different definitions in different regions. The most classic example is the **absolute value function**:

$$|x| = \begin{cases} -x & x < 0 \\ x & x \geq 0. \end{cases}$$

Care has to be taken when evaluating the limits of such functions....

Example

Sketch the function

$$f(x) = \begin{cases} 3 - x, & x \leq 2 \\ \frac{x}{2} + 1, & x > 2 \end{cases}$$

Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$.

Empty and Full Circle Notation:

In the previous sketch, we use the convention that

- ▶ If the end point of a line segment is **not** included in its definition, it terminates with an **open circle**, $\circ$
- ▶ If the end point of a line segment **is** included in its definition, it terminates with an **closed circle** $\bullet$.

Existence of a limit

$\lim_{x \rightarrow a} f(x)$ **exists** if and only if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

So if $\lim_{x \rightarrow a} f(x) = L$ exists, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

though it is not necessary that $f(a) = L$

Example

Sketch the function

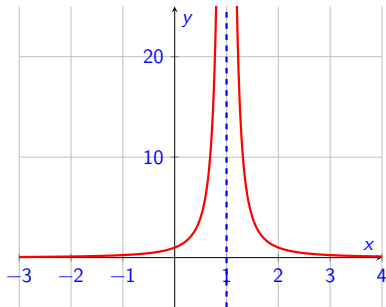
$$f(x) = \begin{cases} 3 - x, & x < 2 \\ 4, & x = 2 \\ \frac{x}{2}, & x > 2 \end{cases}$$

Determine if $\lim_{x \rightarrow 2} f(x)$ exists.

Vertical Asymptotes

Let's revisit an example from earlier:

$$f(x) = \frac{1}{(x-1)^2}$$



Note that the points on the graph having x -coordinates very near to 1 are very close to the vertical line $x = 1$. That is, as x approaches 1, the points on the graph of $f(x)$ are closer to the line $x = 1$.

We call the line $x = 1$ a **vertical asymptote** of the graph.

Vertical Asymptotes

Definition: Vertical Asymptote

The vertical line $x = a$ is a **vertical asymptote** of $f(x)$ if any of $\lim_{x \rightarrow a^-} f(x)$, $\lim_{x \rightarrow a^+} f(x)$, or $\lim_{x \rightarrow a} f(x)$ are ∞ or $-\infty$.

To find a vertical asymptote of a function $f(x) = \frac{p(x)}{q(x)}$, we find a value, a for which $p(a) \neq 0$ but $q(a) = 0$.

Vertical Asymptotes

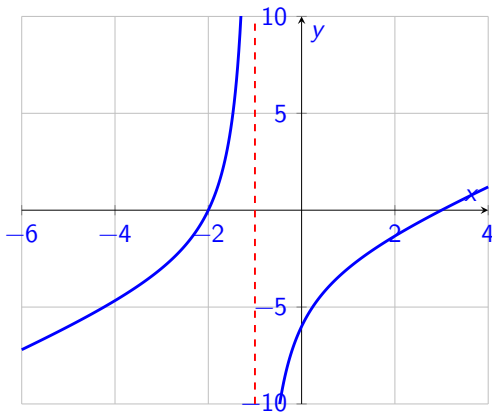
Example

Find any vertical asymptotes of

$$f(x) = \frac{x^2 - x - 6}{x + 1}$$

Vertical Asymptotes

$$f(x) = \frac{x^2 - x - 6}{x + 1}$$



Vertical Asymptotes

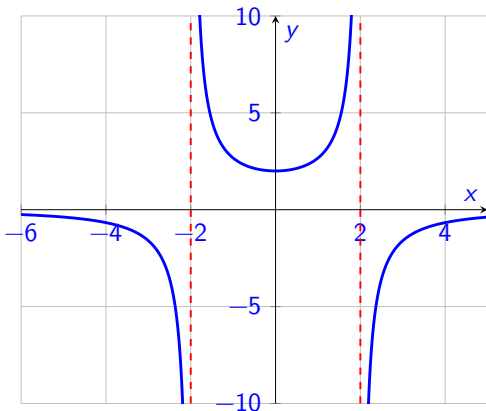
Example

Find all vertical asymptotes of the graph of

$$g(x) = -\frac{8}{x^2 - 4}.$$

Vertical Asymptotes

$$f(x) = -\frac{8}{x^2 - 4}$$



There is a related concept of a **horizontal asymptote**, but we'll save that for later, when we cover “limits at infinity”.

Continuity

Many functions have the property that you can trace their graphs with pen and paper, without lifting the pen from the page. Such functions are called **continuous**.

Some other functions have points where you have to lift the pen occasionally. We say they have a **discontinuity** at such points.

Intuitively, a function is continuous at a particular point if there is no **break** (or “**jump**”) in its graph at that point.

More formally, we define continuity in terms of **limits**.

Continuity

Definition

A function f is **continuous at** $x = a$ if

1. $f(a)$ is defined, i.e., a is in the domain of f ,
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

If $f(x)$ is not continuous at $x = a$ we say it is **discontinuous** at $x = a$.

If f is continuous **at every point** in its domain, we say f **is continuous**.

Many functions are continuous, e.g. all polynomial functions, **most** trigonometric functions (not **tan**), $|x|$, and so on.

Example 1

Determine if $f(x) = \frac{x^2 - 4}{x - 2}$ is continuous at $x = 2$.

Example 2

Determine if $f(x) = \begin{cases} 1 - x & x \leq 0 \\ 2 + x & x > 0 \end{cases}$ is continuous at $x = 0$.

Example 3

Determine if $f(x) = \begin{cases} \frac{\sin(x)}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$ is continuous at $x = 0$.

Example

Consider the function

$$f(x) = \begin{cases} x + 1, & x < 2 \\ bx^2, & x \geq 2 \end{cases}$$

For what value of b is f continuous at $x = 2$?

Example

For what values of x is $f(x) = \frac{2x + 1}{2x - 2}$ continuous?

Exercises

Exercise 3.2.1

Find all the vertical asymptotes of $f(x) = \frac{x+2}{x^2+2x-8}$.

Exercises 3.2.2 (Based on Q1(a), 23/24)

$$\text{Let } g(x) = \begin{cases} 3 & x \leq 0 \\ 2x + 1 & 0 < x < 1 \\ x^2 & x \geq 1. \end{cases}$$

- (i) Sketch the graph of $g(x)$ on the interval $[-3, 4]$, making use of the empty and full circle notation.
- (ii) Compute $\lim_{x \rightarrow 1^-} g(x)$ and $\lim_{x \rightarrow 1^+} g(x)$. Is g continuous at $x = 1$. If not, classify the type of discontinuity.

Exercises

Exercise 3.2.3

For what values of b and c is $f(x) = \begin{cases} x^2 + 1 & x \leq -1 \\ x + b & -1 < x < 1 \\ cx^2 & x \geq 1. \end{cases}$
continuous at $x = -1$ and $x = 1$?

For more practice try the Exercises at the end of
<https://openstax.org/books/calculus-volume-1/pages/2-3-the-limit-laws>, especially Problems 93–103.